x_1	x_2	x_3	y
test	800,000		
test	200,000		

$y = ax_1 + bx_2 + cx_3 + d$

$y = ax_1 + bx_2 + cx_3 + dx_2 + ex_3$

ML Param

- Embedding
- RAG
- → Agent / tool.calls
- → End to Projects
- → Pick a dataset

→ Recommend

→ linear

solve_linear(X, y)

x_1	x_2	x_3	y

solve_linear()

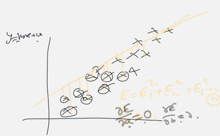
$y = ax_1 + bx_2 + cx_3 + dx_4$

$\frac{\partial E}{\partial a} = 0$ $\frac{\partial E}{\partial b} = 0$ $\frac{\partial E}{\partial c} = 0$ $\frac{\partial E}{\partial d} = 0$

$E = (E_1)^2 + (E_2)^2 + (E_3)^2 + (E_4)^2 + (E_5 \dots)$

$E = 70a^2 - 5a + 7$

y = mx + c



$E = E_1^2 + E_2^2 + E_3^2 + \dots \; \textcircled{n}$

$\frac{\partial E}{\partial a} = 0$, $\frac{\partial E}{\partial b} = 0$.

n column

m-rows

Linear model

$y = \boxed{n+1}$ param..

$y = ax_1 + bx_2 + cx_3 \dots$

$E = E_1^2 + E_2^2 + \dots \; E_m^2$

$\frac{\partial E}{\partial a} = 0$, $\frac{\partial E}{\partial b} = 0$, $\frac{\partial E}{\partial c} = 0$ … $(\boxed{n+1}$ Equ$)$

E

$O(E^3)$

Solving E Equations is rough $O(E^3)$

$O(500^3)$

$O(n+1)^3$

X y

$x_1 \; x_2 \; x_3 \; x_4$ | y_1

$x_{11} \; x_{12} \dots$

solve_lr(X, y)

Poly…

$x \quad x_1^3 \quad x_1^3 \quad x_2^3 \quad x_1^2 \quad x_2^3 \quad x_1 x_2 \quad x_1^2 x_2 \quad x_1 \dots$

IRL

→ Periodic Exe..

y = mx + c

10000 1000

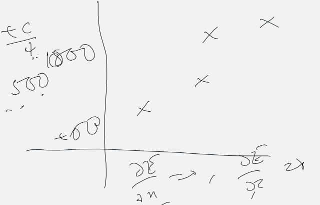
0 500

0.1, 0.2, …

+500

-500

108



$\frac{\partial E}{\partial m}$ → , $\frac{\partial E}{\partial c}$ 2x

$y = ax^2 + bx + c$

10000 1000 100

10^{12}

-1